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RadioLuna

HF/VHF Receiver Payload Feasibility

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1 Executive Summary

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2 Introduction

2.1 RadioLuna Objective

The objective for **RadioLuna** is to deploy micro-satellites in lunar orbit as a constellation of sub-systems, rather than the traditional paradigm of a monolithic satellite containing all possible sensors.

This is possible now as micro-satellite launch capabilities have improved drastically, along with the commoditisation of advanced sensing capabilities. Further, pushing advanced computation and processing of the signals downstream from the satellite will reduce the complexity of each sensor node.

Through this 'modular' approach, **RadioLuna** should help to reduce the cost and risk of any one launch, and encourage a robust academic and commercial lunar ecosystem.

The initial payload proposes the use of a Software-Defined Radio (SDR) receiver with minimal customization. This payload should

- operate as a payload to a small-satellite platform (presumed to be M3 from OHB Italia);
- be sensitive to signals in the 20MHz-200MHz range;
- fit within a 30x30x30cm volume budget; and
- where possible, be made from Commercial Off-the-Shelf (COTS) components;

This payload is intended to be deployed in lunar orbit and will use the dark-side of the moon during full-moons to filter out terrestrial and solar interference to be able to receive the 21cm cosmic hydrogen signal [Pritchard_2012].

2.2 Report Objective

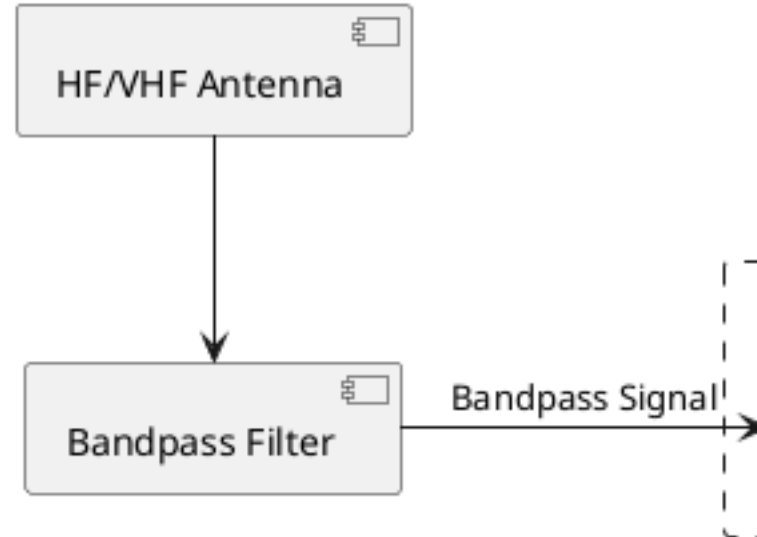
The objective of this report is to develop a systems-level design for a CubeSat payload supporting High/Very-High Frequency (HF/VHF) band communications using a SDR.

SDRs are especially suited to this type of mission as they enable in-orbit configuration without greatly impacting mass or volume of the overall payload. In 2008, the European Space Agency (ESA) defined Software-Defined Payload (SDP)s as payloads "consisting of on-board hardware and software technologies to in-flight reconfigure satellite payloads for multiple different communications scenario"[angeletti:2008].

3 System Design

3.1 Payload Receiver

Talk about digital communications, the SDR, BPF etc.



4 Hardware

4.1 SDR

4.1.1 Requirements

Notes:

- Should be sensitive between 10-100MHz (Confirm)

- Should be COTs

4.2 Antenna

4.2.1 Requirements

- Should be sensitive to 10-100MHz range (Confirm)

- Should be COTs

4.3 Compute Module

4.3.1 Requirements

5 Software

5.1 SDR Software

5.1.1 Requirements

5.2 SDR Controller

5.2.1 Requirements

5.3 Payload Former

5.3.1 Requirements

Disclaimer

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Appendices

A Bill of Materials

#	Vendor	Part #	Name	Function	Leadtime	Qty.	Min. Units	Unit Price	Total
1	Lesker	QF50-200-6X	6-way cross	Chamber	In Stock	1	1	306	306
2	McMaster-Carr	98381A428	Steel Pin	Attach Boat to Flange	In Stock	2	1	3.4	6.8
						3			312.8